

CNC-X100 Safety Guidelines & Operator Standards

Product: CNC Machine | Model: CNC-X100 | Document Type: SAFETY

1. General Operational Safety Rules

Only trained and certified personnel are authorized to operate or service the CNC-X100 machining center.

- Always read and understand all safety placards posted on the machine before operation.
- Maintain a clean, non-cluttered workspace around the machine perimeter (minimum 1.0-meter clearance).
- Never operate machine under influence of medication, drugs, or alcohol.

2. Personal Protective Equipment (PPE) Requirements

All operators and maintenance technicians within the machining bay must wear mandatory PPE:

- Eye Protection: ANSI Z87.1 approved impact-resistant safety glasses with side shields.
- Footwear: Steel-toe safety shoes with slip-resistant soles.
- Hearing Protection: Noise-reduction earplugs or earmuffs when operating near cutting cycles (>85 dBA).

WARNING / SAFETY PRECAUTION:

PROHIBITED NEAR MACHINE: Loose clothing, ties, neck chains, rings, bracelets, and long un-tied hair are strictly prohibited near rotating spindle or chip conveyor.

3. Electrical Safety & High-Voltage Standards

The CNC-X100 operates on high-voltage 400V 3-phase electrical power.

- Electrical cabinet doors must remain closed and key-locked during normal operation.
- Only qualified industrial electricians may open electrical panels or perform electrical troubleshooting.
- Always test electrical terminals for Zero Energy State using calibrated multimeter prior to touching wiring.

4. Moving-Part Hazards & Enclosure Door Interlocks

The machine enclosure features dual interlocked safety doors certified to ISO 13849-1 Category 4 / PL e:

- Door Interlock: Automatically locks enclosure doors while spindle is

rotating or axes are moving in AUTO mode.

- Never bypass, defeat, or short-circuit safety door limit switches under any circumstances.
- Beware of pinch points around automatic tool changer arm and chip conveyor bucket.

5. Emergency Stop (E-Stop) & Lockout / Tagout (LOTO)

Emergency Stop Procedure:

Press red mushroom E-Stop button immediately upon collision, abnormal noise, or fire. Axis drive power is killed instantly and spindle mechanical brake engages within 0.5 seconds.

Lockout / Tagout (LOTO) Procedure:

Prior to entering enclosure for cleaning or maintenance, turn main electrical disconnect to OFF, attach personal padlock and warning tag, and lock pneumatic shutoff valve.